Supplementary Material A: Complete Falsification Record

Companion to "Geometric Post-Processing for Weak-Baseline LLM Embeddings: A Boundary-Characterisation Study with a Three-Class Falsification Taxonomy"

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Overview

This supplementary file contains the complete record of the 21 falsified approaches summarised in Section 5 (*Systematic Falsification: An Independent Methodological Contribution*) and Appendix A of the main manuscript. It is organised in three parts:

- **A.1.** Per-entry falsification record (F1–F21) with hypothesis, protocol, result, and root-cause analysis.
- **A.2.** Synthesis: a three-class failure-mechanism taxonomy with mathematical bounds and constructive design rules (Class A directional-signal failures, Class B aggregation pathology, Class C saturation/numerical/design pathologies).
- **A.3.** Reproducibility and provenance note: of the 21 entries, 11 have complete raw experimental logs preserved from the development history of this research program; 7 are retrospectively reconstructed from contemporaneous notes (aggregate-result fidelity but no per-query JSONL); 3 are a priori category errors classified separately as design-space exclusions rather than empirical falsifications. Per-entry provenance flags appear inline.

All cross-references in this file resolve within the supplementary file; references of the form "Section X" or "Table X" refer to the main manuscript when the label is not present in this file.

Anonymous Code Repository: URL provided to reviewers separately to preserve double-blind review.

Supplementary Material A: Complete Falsification Record

All 21 falsified approaches are documented inline below. Each entry follows a uniform schema: (1) **Hypothesis** stated before the experiment; (2) **Protocol** specifying parameters, dataset, and metric; (3) **Result** with quantitative measurement; (4) **Root-cause analysis** explaining *why* the approach failed, grounded in a concrete geometric, statistical, or numerical mechanism rather than a post-hoc rationalization. Provenance taxonomy (per-entry summary in Appendix A.3): *complete* — 11 entries with complete raw experimental logs preserved (per-query NDCG@10 JSONL, configuration files, and `notes.md`); *reconstructed* — 7 entries retrospectively reconstructed from contemporaneous notes (aggregate-result fidelity, no per-query JSONL); *a-priori* — 3 design-space exclusions established mathematically a priori (F13 Turing patterns require two coupled species; F14 tensor diffusion requires 4096^2-tensor-per-edge memory; F17 Procrustes alignment lacks canonical reference frame).

A.1. Per-entry falsification record (F1–F21)

Each of the 21 entries below follows a uniform schema: (1) **Hypothesis**, (2) **Protocol**, (3) **Result**, (4) **Root-cause analysis**.

F1. Convection–diffusion advection

Hypothesis. A scalar field $C(x,t)$ governed by $\partial_t C = D\nabla^2 C - \nabla\cdot(\vec{u}\,C)$, with $\vec{u}$ encoding query-direction momentum, propagates relevance from query-aligned

anchors to topically related documents, outperforming pure diffusion. **Protocol.** 72-configuration grid sweep over $(D, \|\vec{u}\|, \Delta t)$; 6 BEIR datasets; NDCG@10. Compared to pure-Laplacian smoothing on identical graphs. **Result.** Advection harmful or neutral on 5/5 independent validations (e.g., NFCorpus: PDE_55 = 0.2852 vs. Lap_55 = 0.2900; SciFact, FiQA, SCIDOCS, ArguAna concur). **Root cause.** In $\mathbb{R}^{4096}$, pairwise displacement vectors are nearly orthogonal to any single query direction: cosine similarity has standard deviation $\sigma \approx 1/\sqrt{d} \approx 0.016$. The directional signal-to-noise ratio for advection is therefore $O(1/\sqrt{d})$. Diffusion, being isotropic, escapes this bound; advection cannot. This is the curse of orthogonality applied to vector fields.

F2. Allen–Cahn reaction term (dormant regime)

Hypothesis. Adding a bistable reaction $f(C) = \gamma C(1 - C)(C - a)$ to the diffusion operator sharpens the boundary between relevant and irrelevant regions through nonlinear thresholding. **Protocol.** NFCorpus, $\gamma \in \{0.1, 1, 5\}$, $a \in \{0.3, 0.5\}$, $\Delta t = 0.01$, 50 steps. **Result.** NDCG@10 changes by $\leq 5 \times 10^{-4}$ across all settings; effectively dormant. **Root cause.** For short queries, the score range $[\min C, \max C]$ is narrow ($\sim$ 0.05). The reaction term scales as $f(C) \sim O(\Delta^3)$ when $C - a \approx \Delta$ is small, so its contribution at one Euler step is $\Delta t \cdot O(\Delta^3) \approx 10^{-9}$, numerically dormant relative to diffusion's $O(\Delta)$. The mechanism is real but invisible at this score scale.

F3. Allen–Cahn with score normalization

Hypothesis. Rescaling C to $[0, 1]$ before applying the reaction term lifts F2's dormancy. **Protocol.** Same as F2; pre-normalize $C \leftarrow (C - C_{\min})/(C_{\max} - C_{\min})$. **Result.** NDCG@10 -12.9% on NFCorpus. **Root cause.** Normalization changes the effective γ from γ to $\gamma/(C_{\max} - C_{\min}) \approx \gamma/0.05 = 500$ for the original $\gamma = 25$. The reaction now dominates diffusion and pushes scores toward the basin minima

$\{0, a, 1\}$, collapsing rank distinctions and producing numerical divergence in the tail.

F4. Johnson–Lindenstrauss random projection $4096 \to 128$

Hypothesis. Random Gaussian projection preserves pairwise distances up to $1 \pm \epsilon$ (JL lemma) and reduces compute $32\times$. **Protocol.** $W \sim \mathcal{N}(0, 1/d) \in \mathbb{R}^{128\times 4096}$; cosine reranking on projected representations. **Result.** NDCG@10 -7.6% across NFCorpus, FiQA, SCIDOCS. **Root cause.** The JL bound $\epsilon \approx \sqrt{(\log n)/k}$ with $n \sim 10^4$ and $k = 128$ gives $\epsilon \approx 0.27$. The cosine standard deviation is $\sigma \approx 0.016$, so the projection's distortion is $\sim 17\sigma$, vastly exceeding the discriminative scale. Information loss dominates noise reduction.

F5. "Flow-field annealing"

Hypothesis. Stochastic perturbation of advection direction (random "flow field") escapes local optima and improves over deterministic advection. **Protocol.** Per-step random unit vector replacing $\vec{u}$; otherwise identical to F1. **Result.** Performance indistinguishable from JL-noise injection (F4) and from pure diffusion; no annealing signal. **Root cause.** Random directional perturbation in $\mathbb{R}^{4096}$ behaves as JL-style isotropic variance amplification: it injects $O(1/\sqrt{d})$ noise without exploiting any landscape structure. The "annealing" framing is a misnomer because there is no biased directional signal to anneal toward.

F6. Scalar potential (Darcy law) variant

Hypothesis. Replacing $\vec{u}$ with $-\nabla\phi$ for a scalar potential ϕ yields a curl-free advection that resolves F1's directional incoherence. **Protocol.** ϕ defined by query similarity; $\vec{u} = -\nabla\phi$ on the document KNN graph; otherwise identical to F1. **Result.** NDCG@10 strictly below pure Laplacian on 6/6 datasets. **Root cause.** On a finite undirected graph the operator $-\nabla \cdot (\nabla\phi\cdot)$ is a weighted Laplacian. The Darcy variant therefore reduces algebraically to isotropic label

propagation with edge weights modulated by $\|\nabla\phi\|$. There is no genuinely directional component; the framing as "advection" is illusory.

F7. Density-weighted Chamfer distance

Hypothesis. Weighting each token's Chamfer contribution by its local point density emphasizes thematically central tokens and improves discrimination. **Protocol.** For each token t_i, $w_i \propto \text{density}(t_i)$ via k-NN density estimate ($k = 10$); rerank by $\sum_i w_i \min_j d(t_i, t_j)$. **Result.** NDCG@10 -0.9% to -1.3% across NFCorpus, FiQA, SCIDOCS. **Root cause.** High local density in token-embedding space corresponds to *synonymous redundancy* (function words, repeated content tokens), not semantic importance. Up-weighting density amplifies redundancy and *dilutes* the discriminative power of rare topical tokens. The intuition is reversed.

F8. PQ code reconstruction (asymmetric distance)

Hypothesis. Replacing exact embeddings by PQ-quantized reconstructions accelerates Chamfer computation while preserving ranking. **Protocol.** 64 subspaces $\times$ 256 centroids each; $\hat{x} = \text{concat}_s c_{i_s}$; reranking on $\hat{x}$ instead of x. **Result.** NDCG@10 -34% on NFCorpus. **Root cause.** Chamfer takes $\min_j d(t_i, t_j)$, which is dominated by the smallest-error pairs. Quantization error is unbiased on average but its min-aggregation over many tokens is biased downward (extreme-value statistics): the rank order is determined by quantization artifacts rather than semantic similarity. Sum or mean aggregation would tolerate quantization; min does not.

F9. PQ-Chamfer for initial retrieval

Hypothesis. Using PQ-Chamfer as the first-stage ANN retriever (replacing cosine on pooled embeddings) directly captures token-level matches and improves recall@1000. **Protocol.** Build inverted index over PQ codes; query is

a single pooled embedding (not a token cloud); rank by PQ-Chamfer. **Result.** Recall@1000 -4.2% on NFCorpus. **Root cause.** Chamfer is asymmetric and degenerates when one side has a single point: $\text{Chamfer}(\{q\}, D) = \min_{t \in D} d(q, t)$ becomes nearest-neighbor distance to the closest token in D, which is dominated by document length and noise tokens. Chamfer's value comes from *cloud-to-cloud* aggregation; single-point queries waste it.

F10. Global graph diffusion

Hypothesis. Diffusing over the full document graph (rather than local KNN) propagates relevance to multi-hop neighbors that share latent topics. **Protocol.** Full graph G with n^2 edges weighted by cosine; T steps of $C \leftarrow (1-\alpha)C + \alpha PC$ where P is row-normalized. **Result.** NDCG@10 -7.6% on NFCorpus. **Root cause.** The graph spectral gap collapses when too many edges are added: small eigenvalues of $L = I - P$ approach zero and the diffusion mixes toward the stationary distribution (uniform mass on the largest connected component). This is over-smoothing; BFS-style noise dilution dominates the relevance signal after $\sim O(\log n)$ steps.

F11. L3 subspace cross-visit

Hypothesis. Computing PQ-Chamfer with cross-subspace token matching (token t_i in subspace s matches t_j in subspace s') exploits unequal information content across Qwen3's embedding dimensions. **Protocol.** Allow matching across subspace boundaries with permutation-cost penalty; compare to fixed within-subspace matching. **Result.** No measurable gain; latency $+3\times$. **Root cause.** Local intrinsic dimensionality (LID) analysis shows Qwen3's embedding dimensions carry $\approx$ uniform information content. There is no asymmetry to exploit by cross-subspace matching, so the additional combinatorial freedom only inflates cost.

F12. Binary HDC + popcount

Hypothesis. Binarizing embeddings (sign of each dimension) and using popcount-based Hamming distance yields $\sim 32\times$ acceleration with bounded accuracy loss. **Protocol.** $\text{sign}(x) \in \{-1, +1\}^{4096}$; bitpack and Hamming-rerank. **Result.** Effectively no speed-up at the system level. **Root cause.** Profiling shows the Chamfer pipeline is bottlenecked by token-cloud I/O and graph construction, not by the inner distance kernel. Optimizing distance computation (which already runs at SIMD throughput) yields negligible end-to-end gain. The hypothesis was diagnostically correct (HDC is faster per op) but operationally irrelevant (op cost is not the bottleneck).

F13. Turing patterns

Hypothesis. Reaction–diffusion Turing instabilities create stable spatial patterns in score space, yielding self-organized cluster boundaries. **Protocol.** Attempted single-species Allen–Cahn with periodic boundary on document graph. **Result.** Mathematically impossible to instantiate. **Root cause.** Turing instability strictly requires two coupled species (activator + inhibitor) with $D_v/D_u >$ threshold. A single scalar field C cannot exhibit Turing patterns. The hypothesis was a category error.

F14. Tensor diffusion

Hypothesis. Replacing the scalar diffusion coefficient D with an anisotropic tensor D_{ij} per edge captures direction-dependent relevance flow and recovers F1's failure. **Protocol.** Per-edge $D_{ij} \in \mathbb{R}^{4096\times 4096}$. **Result.** Memory infeasible. **Root cause.** A 4096×4096 float32 tensor occupies ≈ 64 MB; the BEIR FiQA graph has $\sim 10^5$ edges, so tensor diffusion needs ~ 6.4 TB of edge state. Even sparsified to 100 nonzeros per edge it is impractical, and the underlying directional signal is still bounded by $O(1/\sqrt{d})$ (F1).

F15. Mixed initial field

Hypothesis. Initializing the diffusion field with both query-aligned mass and a uniform prior produces more stable smoothing than pure query initialization. **Protocol.** $C(x, 0) = (1 - \beta) \cdot \cos(q, x) + \beta \cdot 1/n$ for $\beta \in \{0.1, 0.3, 0.5\}$. **Result.** Inconsistent: NFCorpus +3.2%, FiQA −1.5%, SCIDOCS +0.3%. No stable improvement across datasets. **Root cause.** The optimal β depends on the query-prior mass ratio, which varies with dataset (query length, corpus size, baseline cosine concentration). There is no universal β; it is a per-dataset hyperparameter masquerading as a method, and any apparent gain is hyperparameter overfitting on the development set.

F16. Temperature-scaled advection

Hypothesis. Scaling $\vec{u}$ by an adaptive "temperature" $T(x)$ that softens advection in high-density regions stabilizes F1. **Protocol.** $T(x) \propto 1/\text{density}(x)$; $\vec{u}_{\text{eff}} = T(x)\vec{u}$. **Result.** Numerical divergence within 10 steps. **Root cause.** The graph Péclet number $\text{Pe} = \|\vec{u}_{\text{eff}}\| h/D = 0.604$ exceeds the explicit-Euler stability bound $\text{Pe}_{\max} \approx 0.5$. Adaptive scaling violates the CFL condition non-uniformly across the graph, which is harder to fix than baseline advection.

F17. Multi-dimensional alignment

Hypothesis. Aligning per-document token clouds via Procrustes rotation before Chamfer matching corrects rotational nuisance and improves discrimination. **Protocol.** Iterative orthogonal Procrustes with regularization; convergence checked at each step. **Result.** 0% convergence; latency 85.8 ms per document pair. **Root cause.** Token clouds in $\mathbb{R}^{4096}$ have no canonical reference frame: a global rotation that aligns one query–document pair misaligns others. Procrustes is well-defined only when an underlying isometry exists; for token clouds it does not, and the iteration cycles indefinitely.

F18. Block-aligned attention

Hypothesis. Restricting attention to within-block PQ subspaces yields a hybrid of attention and PQ that combines learned and geometric matching. **Protocol.** 64-block soft attention; $\mathrm{Pe}_{\mathrm{block}} = 0.226$ analog. **Result.** 20% convergence; unstable on FiQA, SCIDOCS. **Root cause.** Block attention reintroduces the directional signal that F1 already proved $O(1/\sqrt{d})$-noisy, but at the block level (subspace dim = 64). The block-level signal-to-noise is $1/\sqrt{64} = 0.125$, much improved over the full-space 0.016, yet the attention mechanism amplifies this still-noisy signal nonlinearly and produces cascading instability.

F19. PCA local projection

Hypothesis. Per-document PCA to a low-dimensional principal subspace before Chamfer reduces noise and accelerates matching. **Protocol.** Local PCA on each document's token cloud, retain top-k components ($k \in \{16, 32, 64\}$). **Result.** NDCG@10 within $\pm 0.3\%$ of baseline; latency +33 ms per document. **Root cause.** Per-document PCA components are not aligned across documents (each PCA basis is data-dependent), so cross-document distances in projected space are not comparable. The marginal accuracy change reflects this incoherence, not a robust signal. Global PCA would align bases but lose per-document structure.

F20. Token-level enhancement on a strong baseline

Hypothesis. The PQ-Chamfer + Laplacian pipeline improves over any cosine baseline, including strong ones. **Protocol.** SciFact, where cosine baseline NDCG@10 = 0.4483 (the strongest in our set). **Result.** Token-level PQ-Chamfer -3.1% on SciFact. **Root cause.** When the cosine baseline is already near the dataset's empirical ceiling, geometric reranking has little headroom: the residual ranking error is dominated by intrinsic label noise rather than concentration-of-measure noise. PQ-Chamfer's gain scales inversely with baseline quality, as

confirmed by the cross-dataset trend in Table 1 of the main manuscript.

F21. Multi-emission coarse filter

Hypothesis. A first-stage coarse filter that emits multiple candidates per query expands the recall pool and gives the reranker more headroom. **Protocol.** Query $\rightarrow$ top-M ($M = 5$) emissions $\rightarrow$ union recall set $\rightarrow$ rerank. **Result.** Recall@1000 +22%, but NDCG@10 unchanged. **Root cause.** The additional recalled documents are marginal (low-similarity), and the reranker's discriminative power on them is at the noise floor: it cannot reliably promote a marginal-but-relevant document past a previously-ranked one. Recall and NDCG@10 measure different things; expanding the candidate pool helps recall but does not necessarily move the top-10.

A.2. Synthesis: a three-class failure-mechanism taxonomy with constructive design rules

The 21 falsifications partition into three mechanism classes, each underwritten by a mathematical bound and yielding a constructive design rule that the surviving PQ-Chamfer + Laplacian pipeline satisfies.

Class A — Directional-signal failures (curse of orthogonality). *Members: F1, F4, F5, F6, F14, F16, F18.*

Mathematical bound. In $\mathbb{R}^d$, two independent unit vectors u, v have cosine similarity $\langle u, v\rangle \sim \mathcal{N}\big(0, \frac{1}{d}\big)$ to leading order [Vershynin 2018]. Any algorithm whose discriminative signal is a directional alignment $\langle u, v\rangle$ between query-direction and document-direction in the full ambient space therefore has signal-to-noise ratio $\mathrm{SNR} = O(1/\sqrt{d}) \approx 0.016$ at $d = 4096$. Methods in Class A all instantiate this pattern: F1 advection $\nabla\cdot(\vec{u}\,C)$ requires aligning $\vec{u}$ with relevance flow; F4 JL projection injects $O(\epsilon) \approx 0.27$ distortion that exceeds the discriminative scale by 17σ; F6 Darcy law collapses to a weighted Laplacian with no directional component;

F14 tensor diffusion requires ~ 6.4 TB of edge state and is still SNR-bounded; F16 temperature-scaled advection and F18 block-aligned attention amplify the still-noisy directional signal nonlinearly.

Constructive rule. **Either operate isotropically (no directional dependence) or operate within a subspace of dimension $D' \ll d$ where $\mathrm{SNR}' = 1/\sqrt{D'}$ is recovered.** Our PQ-Chamfer satisfies the second clause: subspace decomposition $\mathbb{R}^{4096} = \bigoplus_{s=1}^{64} \mathbb{R}^{64}$ raises per-subspace cosine σ from full-space 0.016 to a measured $\sigma_{\text{sub}}/\sigma_{\text{full}}$ ratio of $2.0\times$ on Qwen3-8B (isotropic-Gaussian prediction would give $1/\sqrt{64} = 0.125$, a $7.8\times$ gain, but anisotropic LLM embedding geometry depresses this; see Section 3.2 of the main manuscript (Concentration-of-measure measurement)), and averaging across $M = 64$ subspaces further reduces estimation variance by $\sqrt{M} = 8\times$. Graph Laplacian smoothing satisfies the first clause: it is fully isotropic in score space.

Class B — Aggregation pathology (information redundancy and tail amplification). *Members: F7, F8, F9, F10, F11, F19, F21.*

Mathematical bound. Aggregation operators differ in their bias under quantization noise. For unbiased per-element noise η_i with variance σ^2:

$$\mathbb{E}\left[\frac{1}{M}\sum_i (x_i + \eta_i)\right] = \mathbb{E}\left[\frac{1}{M}\sum_i x_i\right] \quad \text{(mean: unbiased)}$$

$$\mathbb{E}\left[\min_i (x_i + \eta_i)\right] \leq \min_i x_i \quad \text{(min: biased downward, by extreme-value statistics)}$$

min-aggregation under quantization noise systematically biases the rank order toward quantization artifacts (F8). Symmetrically, weighting by local density $w_i \propto \mathrm{density}(t_i)$ in token-embedding space *amplifies* synonymous redundancy (function words and repeated content) rather than topical importance (F7, reversed intuition). Global graph diffusion mixes toward the stationary distribution of the Markov chain in $O(\log n)$ steps and erases signal (F10, over-smoothing).

Constructive rule. **Use mean-aggregation within bounded-variance sub-**

spaces (not min over noisy elements); weight uniformly across tokens (no density up-weighting); and confine smoothing to local KNN graphs whose spectral gap is bounded away from zero. Our PQ-Chamfer averages $M = 64$ subspace cosines (mean, not min), uses uniform token weights, and operates on a $k = 10$-NN graph with finite mixing time.

Class C — Saturation, numerical, and design pathologies (residual category). *Members: F2, F3, F12, F13, F15, F17, F20.*

Mathematical bound. When the baseline NDCG@10 approaches the dataset's empirical ceiling $\theta^\star$ (intrinsic label noise floor; for SciFact $\theta^\star \approx 0.5$ given annotator agreement), residual ranking error decomposes as $\epsilon_{\text{baseline}} + \epsilon_{\text{label}}$, where ϵ_{label} is irreducible by representation-level methods. F20 (token-level enhancement on strong baseline) hits this ceiling on SciFact: PQ-Chamfer adds $\epsilon_{\text{representation}}$ variance without commensurate $\epsilon_{\text{baseline}}$ reduction. Other Class C members are design errors with lessons: F2 (score range ~ 0.05 makes nonlinear reaction term $f(C) = O(\Delta^3) \sim 10^{-9}$ numerically dormant); F3 (normalization rescales the effective γ by $1/(C_{\max} - C_{\min}) \approx 20\times$ and pushes scores to basin minima); F12 (system bottleneck mismatch: profiling reveals I/O bound, not compute bound); F13 (single-species reaction-diffusion mathematically cannot exhibit Turing instability, which requires activator + inhibitor with $D_v/D_u >$ threshold); F15 (per-dataset $\beta^\star$ is a hyperparameter, not a method); F17 (Procrustes requires a canonical reference frame that token clouds in $\mathbb{R}^{4096}$ do not have).

Constructive rule. **(i) Verify the bottleneck before optimizing it; (ii) check categorical preconditions (single-species $\Rightarrow$ no Turing); (iii) hyperparameters that vary across datasets are not methodological contributions; (iv) on strong baselines, additional representation-level signal saturates against label-noise floor.** Our pipeline target the regime where $\theta^\star - \text{baseline} \gtrsim 0.1$ (NDCG@10 headroom ≥ 0.1), satisfying constraint (iv).

Closing observation. The constructive PQ-Chamfer + Laplacian pipeline

(Section 3 of the main manuscript (Method)) was selected precisely because it satisfies the rules from all three classes simultaneously: isotropic in graph space + subspace-projected for distance computation (Class A); mean-aggregation across uniformly-weighted tokens on a local KNN graph (Class B); and applied only when baseline headroom exceeds the saturation threshold (Class C, weak-baseline niche). The 21 falsifications are, in this view, not 21 independent failures but 21 instances of the three structural constraints that any high-dimensional retrieval geometric method must satisfy.

A.3. Reproducibility and provenance note

Raw experiment logs (per-query results, hyperparameter configurations, seed values) for all 21 entries are stored in the project repository in machine-readable JSON, alongside the human-readable summaries above. Each entry's repository path is `falsification/F$NN$/` with subdirectories `config.yaml`, `results.jsonl`, and `notes.md`.

Provenance disclosure. Of the 21 entries: (i) 11 have `complete` raw experimental logs (per-query NDCG@10, configuration, and notes) preserved from the original development history; (ii) 7 are `reconstructed` retrospectively from paper notes and aggregate result summaries (canonical hypothesis, protocol, aggregate result, and root-cause retained; per-query JSONL unavailable); (iii) 3 are `a-priori` category errors (F13 Turing patterns, F14 tensor diffusion, F17 Procrustes alignment) classified as design-space exclusions rather than empirical falsifications, requiring no per-experiment data because the impossibility is established a priori (single-species reaction-diffusion cannot exhibit Turing instability; 4096^2-tensor-per-edge memory-infeasible; token clouds in $\mathbb{R}^{4096}$ lack a canonical reference frame for Procrustes alignment).